

| Criteria                     | Virtual machines                                                                | App service                                            | Azure functions                      | Azure Kubernetes service                   | Container instances                                  |
|------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------|--------------------------------------------|------------------------------------------------------|
| Application composition      | Agnostic                                                                        | Applications, containers                               | Functions                            | Containers                                 | Containers                                           |
| Density                      | Agnostic                                                                        | Multiple apps per instance via app service plans       | Serverless                           | Multiple containers per node               | No dedicated instances                               |
| Minimum number of nodes      | <u>1</u>                                                                        | 1                                                      | Serverless                           | <u>3</u>                                   | No dedicated nodes                                   |
| State management             | Stateless or Stateful                                                           | Stateless                                              | Stateless                            | Stateless or Stateful                      | Stateless                                            |
| Web hosting                  | Agnostic                                                                        | Built-in                                               | Not applicable                       | Agnostic                                   | Agnostic                                             |
| Deployment to dedicated VNet | Supported                                                                       | Supported                                              | Supported                            | Supported                                  | Supported                                            |
| Hybrid connectivity          | Supported                                                                       | Supported                                              | Supported                            | Supported                                  | Not supported                                        |
| Local debugging              | Agnostic                                                                        | IIS Express, others                                    | Visual Studio or Azure Functions CLI | Minikube, others                           | Local container runtime                              |
| Programming model            | Agnostic                                                                        | Web and API applications, WebJobs for background tasks | Functions with triggers              | Agnostic                                   | Agnostic                                             |
| Application update           | No built-in support                                                             | Deployment slots                                       | Deployment slots                     | Rolling update                             | Not applicable                                       |
| Autoscaling                  | Virtual machine scale sets                                                      | Built-in service                                       | Built-in service                     | Pod auto-scaling, cluster auto-scaling     | Not supported                                        |
| Load balancer                | Azure Load Balancer                                                             | Integrated                                             | Integrated                           | Azure Load Balancer or Application Gateway | No built-in support                                  |
| Scale limit                  | Platform image: 1000 nodes per scale set, Custom image: 600 nodes per scale set | 20 instances, 100 with App Service Environment         | 200 instances per Function app       | 100 nodes per cluster (default limit)      | 20 container groups per subscription (default limit) |
| Multi-region failover        | Traffic manager                                                                 | Traffic manager                                        | Azure Front Door                     | Traffic manager                            | Not supported                                        |
| SSL                          | Configured in VM                                                                | Supported                                              | Supported                            | Ingress controller                         | Uses sidecar container                               |

Table 5: A comparison of the different services provided by Microsoft Azure (Reference: <https://docs.microsoft.com/en-us/azure/architecture/guide/technology-choices/compute-decision-tree/>)